

FUNDAMENTALS OF STATISTICAL MECHANICS / 1

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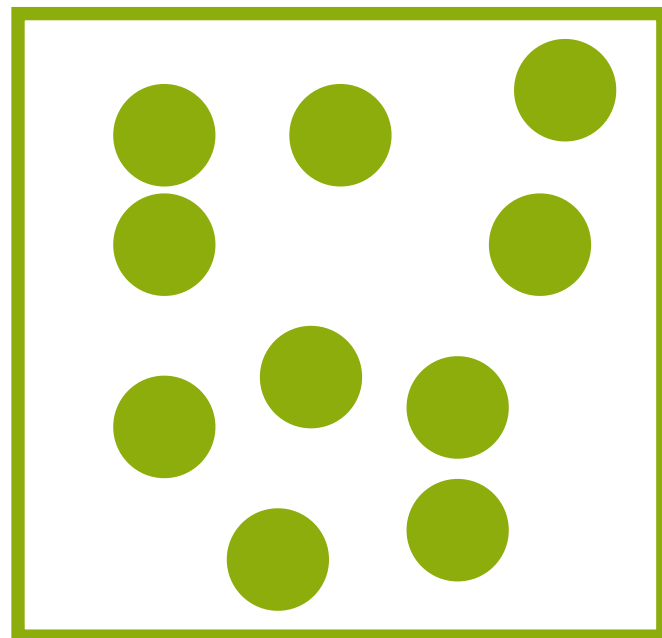
SS 2026

CONTENTS

- Why statistical mechanics?
- Microcanonical ensemble and its probability distribution
- Entropy and the second law of thermodynamics

WHAT IS STATISTICAL MECHANICS?

The idea behind **statistical mechanics**: *it is too complicated (and also redundant) to describe a system composed by $N \gg 1$ particles on a particle-by-particle basis*



Even if we could solve the individual trajectories of all particles, ***this is not the info we want***

We want a **large-scale, serviceable description of the properties of the matter** in the box: density, temperature, pressure ...

Statistical mechanics: from microscopic to macroscopic description, using statistical concepts

Classical physics: number of particles N , temporal evolution of their individual coordinates and momenta based on initial conditions

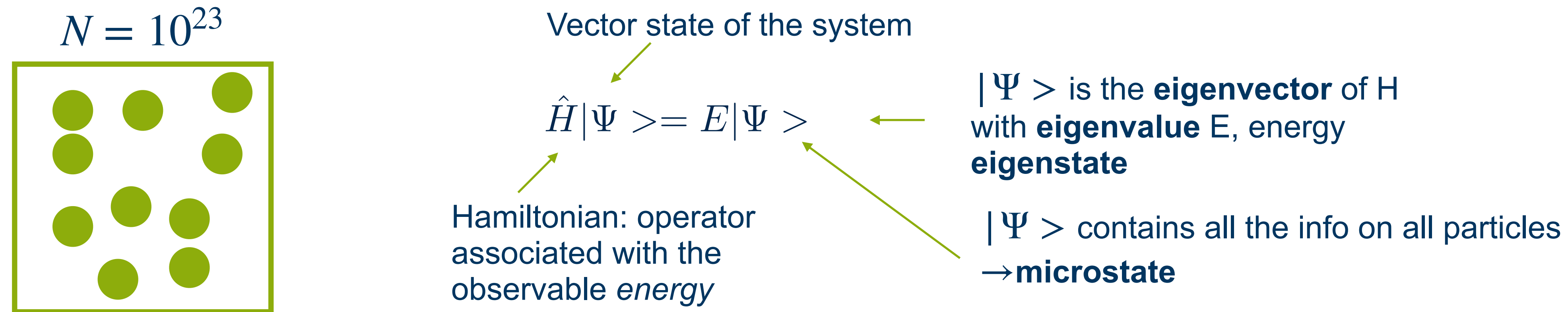
Quantum mechanics: number of particles N , temporal evolution of the wavefunction Ψ based on its initial condition

Amount of gas (moles), pressure, temperature ...
thermodynamics !!!

In this course, we will build the **language** and the **methodology** for this transition

MICROSCOPIC DESCRIPTION IN QUANTUM MECHANICS

Quantum mechanics point of view: **time-independent Schrödinger equation**



System with $N \gg 1$

We cannot/ do not want to calculate the microstate of the system (==we do not want to know the wavefunction Ψ of the system)

We focus on the **observables** (== **macroscopic** quantities of the system)

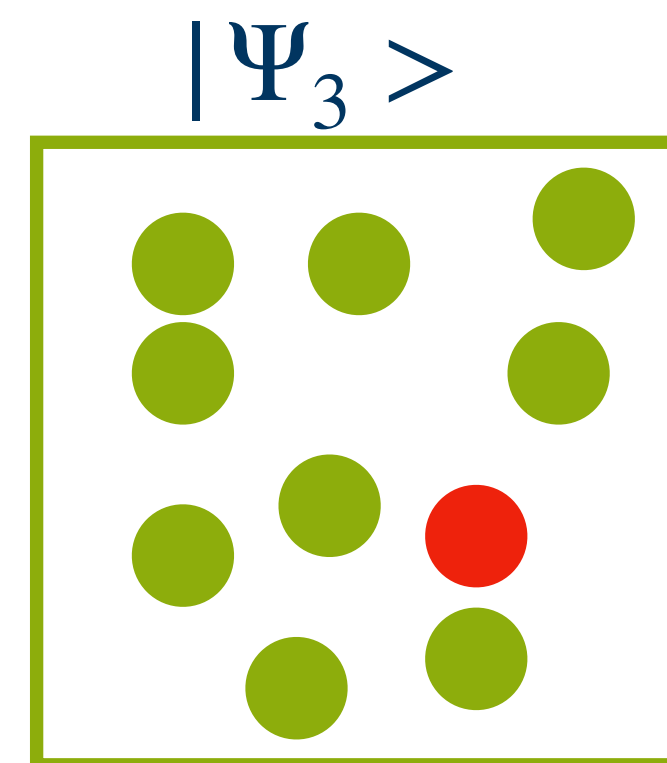
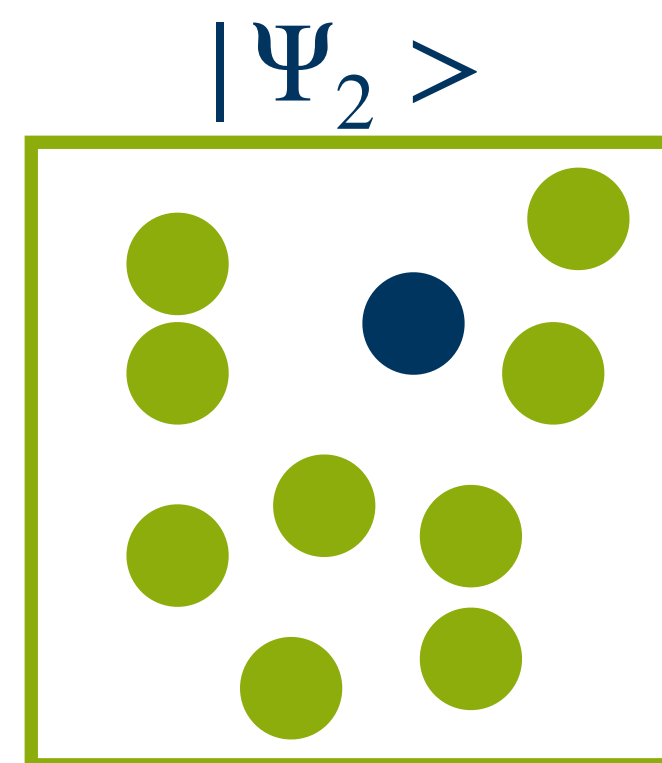
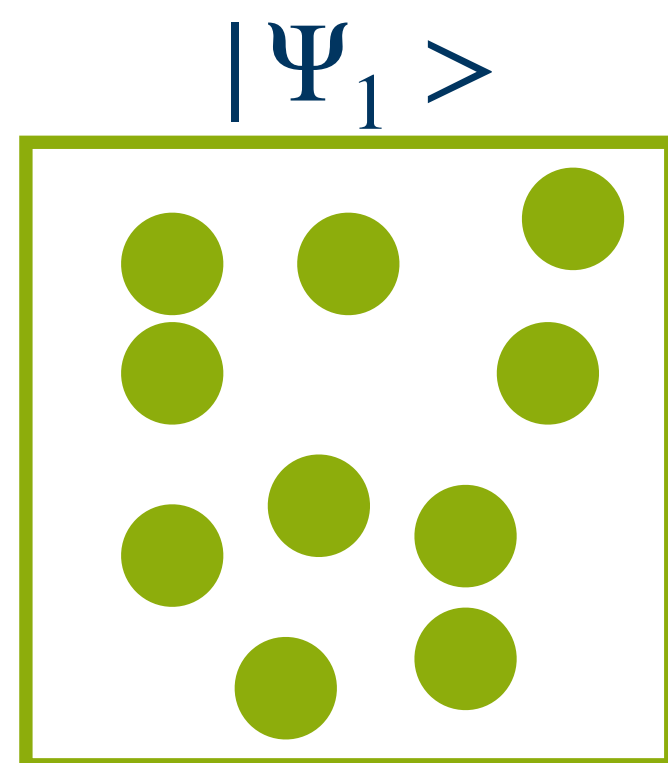
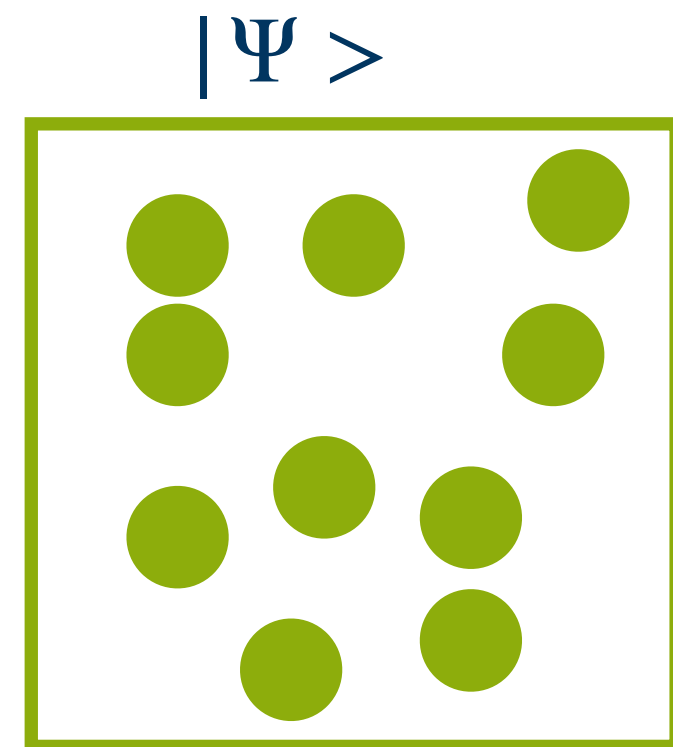
OBSERVABLES: QUANTUM MECHANICS VS STATISTICAL MECHANICS APPROACH

Quantum mechanics: we assume we know the microstate Ψ of the system

Expectation value of the operator $\hat{O}$ associated with the microstate Ψ :

$$\langle \hat{O}_\Psi \rangle = \langle \Psi | \hat{O} | \Psi \rangle \quad \leftarrow \text{Pure state}$$

Statistical physics: let's assume we have a system with energy E subject to small perturbations; perturbations are small enough to keep energy constant; at any time, the system may be in any of a number of microstates $|\Psi_i\rangle$ characterized by the same energy
 → **mixed state**. We define **n**: **basis of mixed states**



Expectation value of operator/ observable in the **mixed state**:

$$\langle \hat{O} \rangle = \sum_n p(n) \langle n | \hat{O} | n \rangle$$

The observables are **macroscopic**

now, we look for an appropriate $p(n)$, probability of “finding” the system in n

PLAN FOR THE FIRST ~3 WEEKS

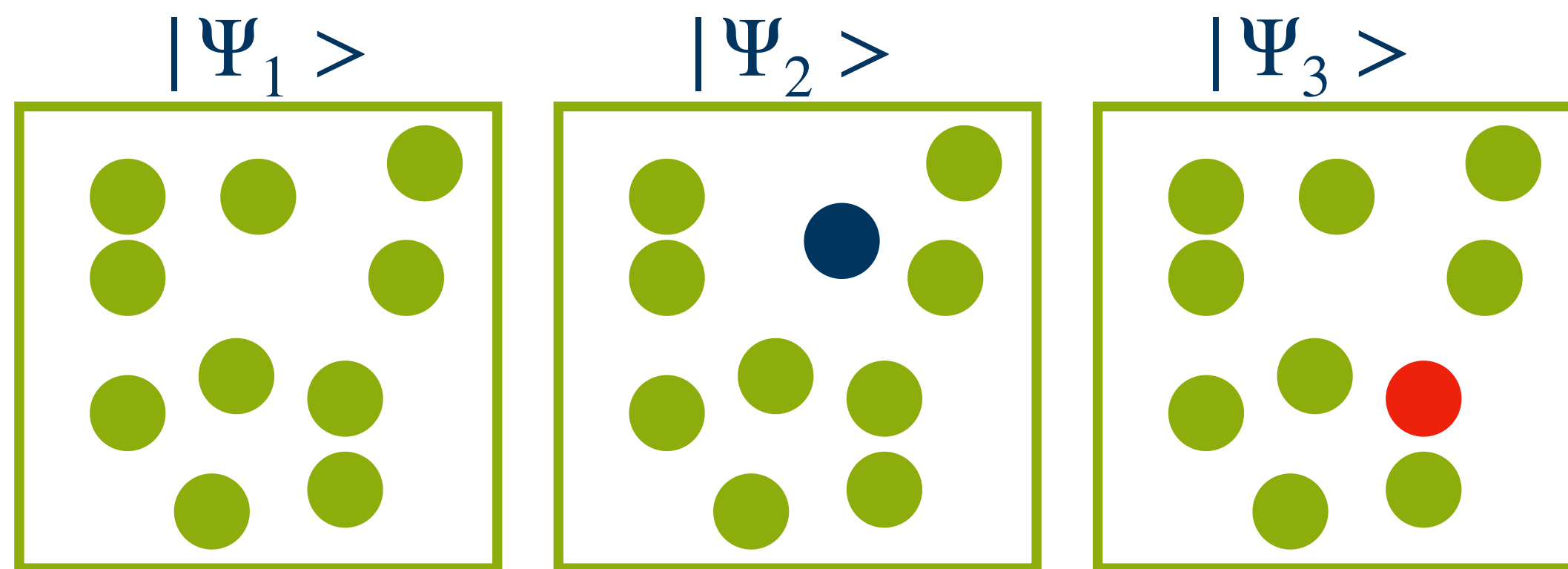
$$\langle \hat{O} \rangle = \sum_n p(n) \langle n | \hat{O} | n \rangle$$

Look for $p(n)$ and a number of quantities we can calculate once we know $p(n)$ in systems which become progressively more complex:

- Microcanonical ensembles: all microstates in the ensemble have the same energy
- Canonical ensembles: energy can change, but same number of particles in each microstate
- Grandcanonical: number of particles can change too

MICROCANONICAL ENSEMBLE

Microcanonical ensemble: isolated (no change in energy or number of particle) system with $N \gg 1$ particles/elements (ensemble), characterized by **fixed energy**, i.e. **all microstates have the same energy**



The system is jogged by perturbations between **many microstates characterized by the same energy**

We assume that our system is

- **isolated**: no energy or particle exchange with the “outside” &
 - **in equilibrium**: energy and momentum in the system have been **redistributed** among the particles, any memory of whatever special initial conditions has been lost
- the probability distribution is independent on time → the **macroscopic observables are independent on time**

PROBABILITY DISTRIBUTION IN MICROCANONICAL ENSEMBLES

Assumption: **for an isolated system in equilibrium, all accessible microstates** (== systems with the same energy) **are equally likely**

$\Omega(E)$: number of microstates with energy E

$$p(n) = \frac{1}{\Omega(E)}$$

Probability that the system with fixed energy E is in a given microstate $|n\rangle$

→ **all accessible microstates are equally likely**

▲ We assume that all the accessible states have constant energy E , so $|d\rangle$ with energy $\neq E$ have $p(d) = 0$

▲ In quantum systems energy levels are discrete. But many particles → energy levels are finely spaced, $\sim$ continuum

Q: what is $\sum_{n=1}^{\Omega(E)} p(n)$?

$$p(n) = \frac{1}{\Omega(E)}$$

Is the probability distribution for the micro canonical ensemble, where all microstates have the same energy

DEFINITION OF ENTROPY



Entropy S of the system: $S(E) = k_B \log \Omega(E)$

Number of microstates with energy E

Important: a number of definition will seem to come out of nowhere; in further lectures we will relate them to macroscopic concepts which we already have e.g. level of information for entropy

Boltzmann's constant: $k_B \sim 1.381 \times 10^{-23} \text{ J K}^{-1}$

Q: what is the entropy of a system composed of N coins, with $S_1 = k_B \log 2$ the 'unit of entropy'?

Two **non-interacting, isolated** systems with energies E_1 and E_2 respectively:

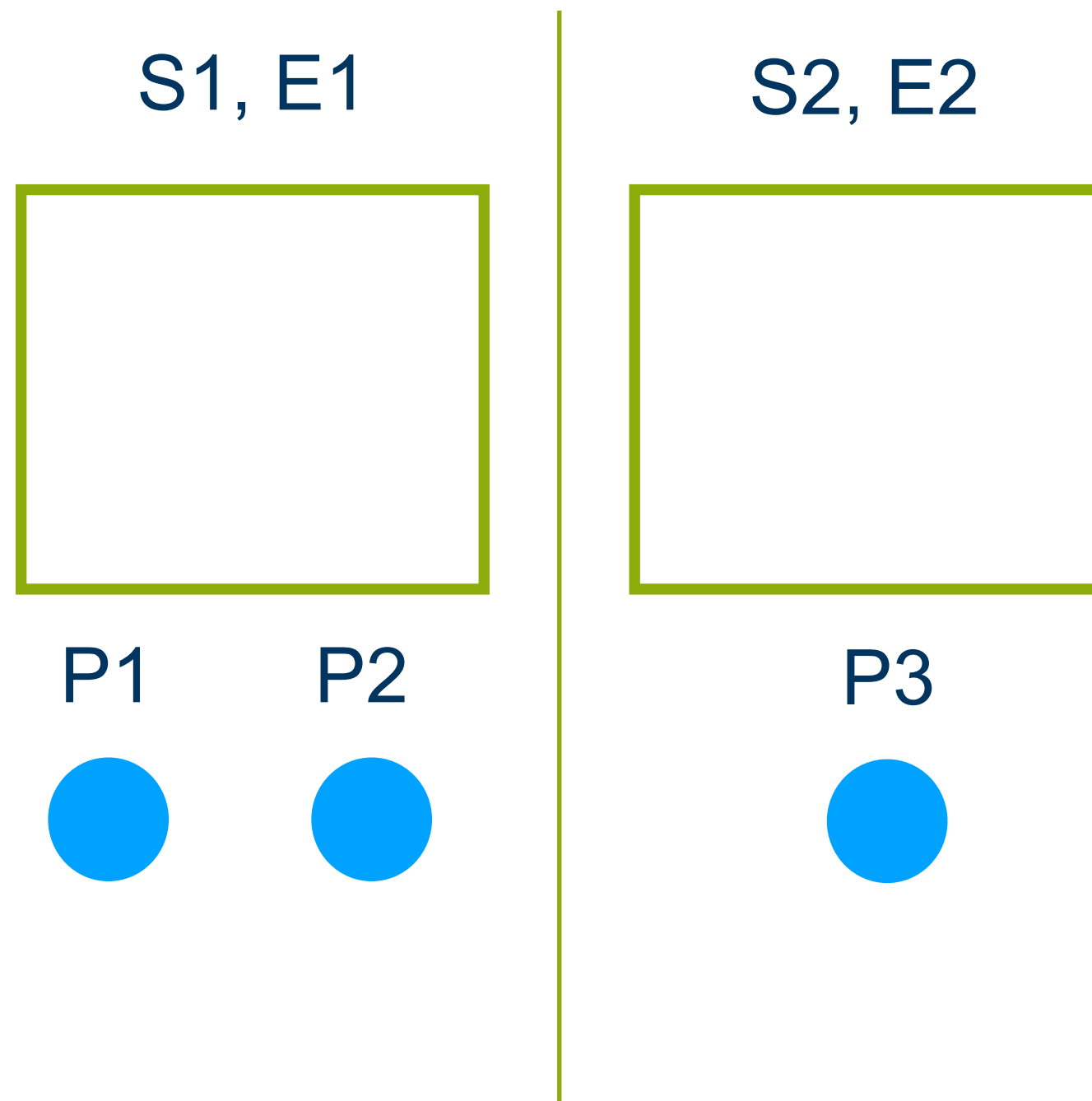
Total number of states: $\Omega(E_1, E_2) = \Omega_1(E_1)\Omega_2(E_2)$

Entropy: $S(E_1, E_2) = k_B \log(\Omega_1(E_1)\Omega_2(E_2)) = k_B \log \Omega_1(E_1) + k_B \log \Omega_2(E_2) = S_1(E_1) + S_2(E_2)$

Entropy is an additive quantity

ENTROPY OF ISOLATED SYSTEMS

Total entropy of the system S composed of the **isolated** systems $S1$ and $S2$ (== $P1$ and $P2$ stay in $S1$, $P3$ stays in $S2$)



Number of microstates for $S1$, with energy $E1$:

$P1 - P2$ 2

$P2 - P1$

Number of microstates for $S2$, with energy $E2$:

$P3$ 1

Number of microstates for the two isolated systems:

$$2 \times 1 = 2$$

Definition of entropy of a system: $S(E) = k_B \log \Omega(E)$ ← Number of microstates with energy E

$$S = k_B \log(2) + k_B \log(1) = k_B \log(2) = k_B \log(2 \times 1)$$

Ok, we recover
← $S(E_1, E_2) = S_1(E_1) + S_2(E_2)$ for **isolated** systems

REMOVING THE PARTITION

Now the systems can communicate: all particles can access all boxes, but **we assume the energy level of each individual box remains the same** (== if all particles have the same energy, 2 particles in S1, 1 particle in S2)

$$S = E_1 + E_2$$

S1, E1

S2, E2



P1

P2

P3



Number of microstates for communicating S1 & S2,
with fixed energy levels

(S1) P1 - P2 (S2) P3

(S1) P2 - P1 (S2) P3

(S1) P3 - P1 (S2) P2

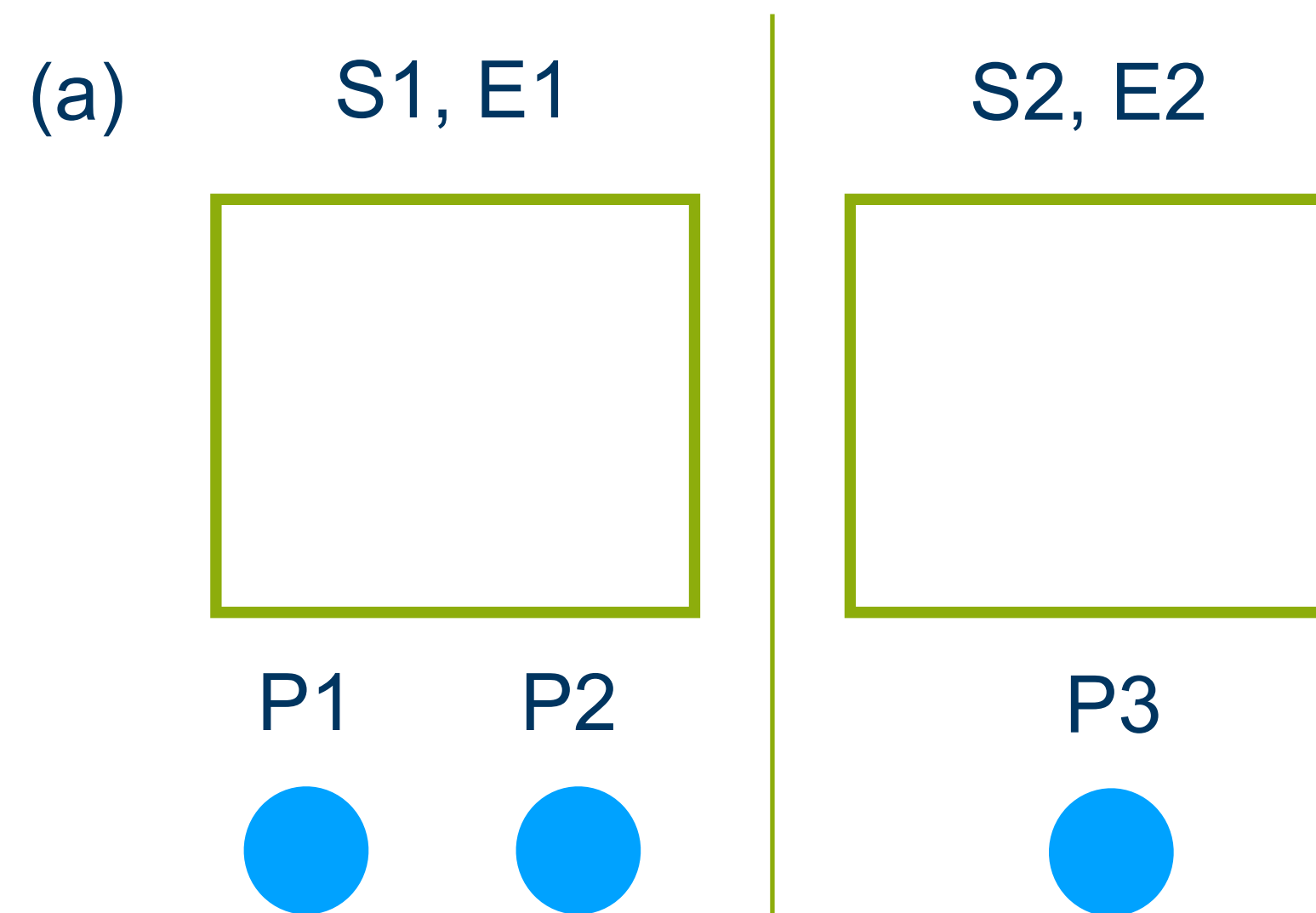
(S1) P1 - P3 (S2) P2

(S1) P2 - P3 (S2) P1

(S1) P3 - P2 (S2) P1

Removing the partition, **more microstates are accessible
then before**

SECOND LAW OF THERMODYNAMICS

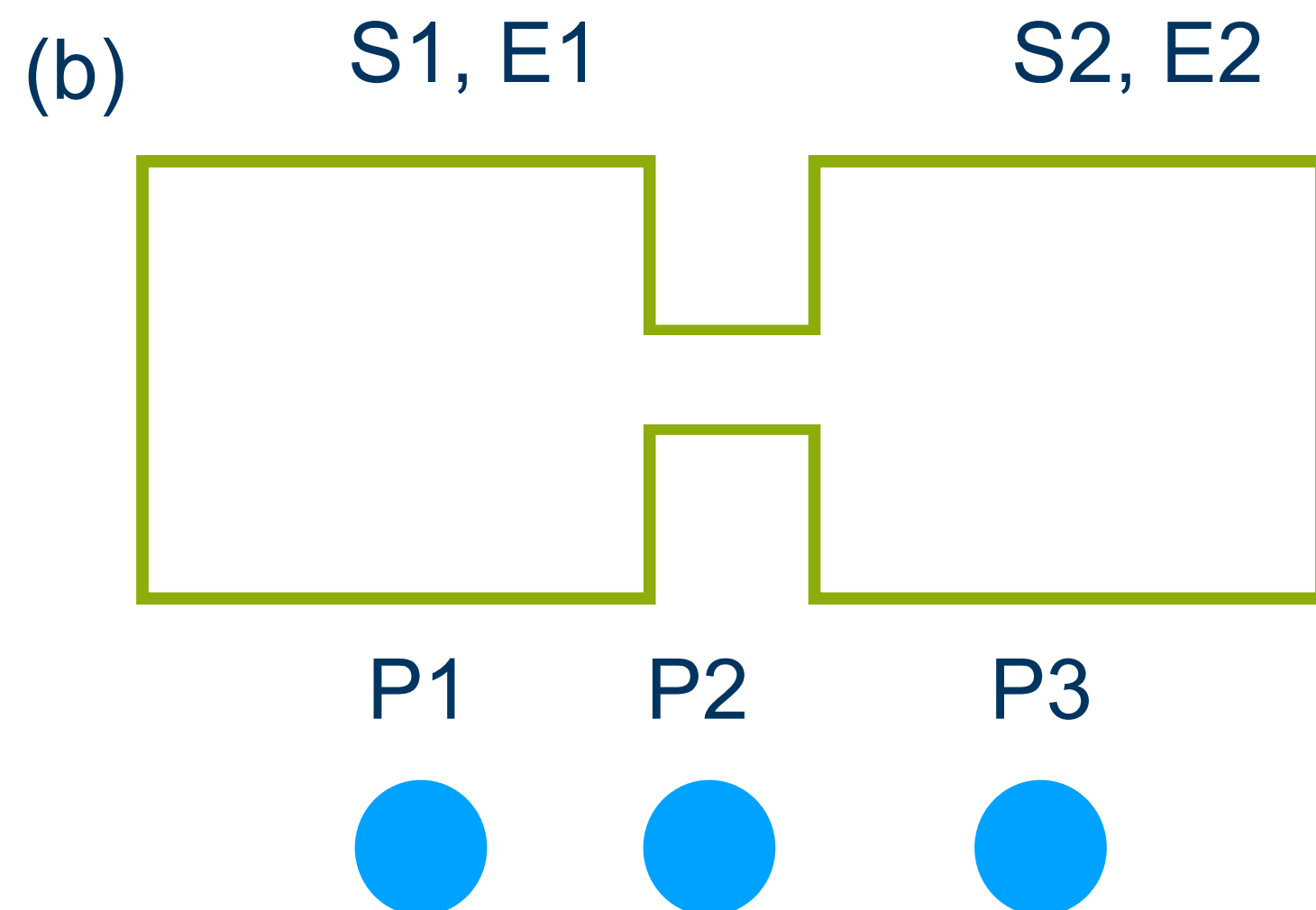


(a) isolated systems

$$S = S_1(E_1) + S_2(E_2) = k_B \log(\Omega_1 \times \Omega_2)$$

(b) systems communicate

$$S = k_B \log(\Omega) \geq k_B \log(\Omega_1 \times \Omega_2) = S_1(E_1) + S_2(E_2)$$



Second law of thermodynamics: when constraints on a system are removed (or equivalently, when previously isolated systems are brought together), the **total number of available states increases** → **entropy increases**

$$S \geq S_1(E_1) + S_2(E_2)$$

SUMMARY

- **Microstate:** specific configuration of a system at the microscopic level (== what each particle is doing: wavefunction in QM, known position and momentum particle-by-particle in classical physics)
- **Macrostate:** macroscopic description of a system (e.g.: values for energy, temperature, volume, pressure, ...)
- **Microcanonical ensemble:** isolated system with $N \gg 1$ elements/ particles (ensemble) characterized by fixed energy, with all microstates equally likely
- **Entropy:** (for now) measure of number of possible microstates in a system, $S = k_B \log \Omega$
- **Second law of thermodynamics:** entropy increases

References: Tong, Sec. 1.1 to 1.2.1